Modulus Functions & the Factor/Remainder Theorem

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational] • Target: Solving Modulus Equations**

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. What are the coordinates of point A? Show your complete algebraic work.

Final Answer: (2, 0). To find the x-intercepts of the function $f(x) = x^2 - 6x + 8$, we need to solve for x when $f(x) = 0$. This means we need to find the roots of the quadratic equation $x^2 - 6x + 8 = 0$. We can factor this quadratic equation. We are looking for two numbers that multiply to 8 and add to -6. These numbers are -2 and -4. Therefore, the equation can be factored as $(x - 2)(x - 4) = 0$. Setting each factor to zero gives us the x-intercepts: $x - 2 = 0 \Rightarrow x = 2$ and.

2 **[Foundational] • Target: Modulus Inequalities**

The diagram shows the graph of the function $y = f(x)$. The function is defined by $f(x) = x^2 + bx + c$. The graph passes through the point $(1, 0)$ and has a vertex at $(2, -1)$. Find the value of b.

Final Answer: -4. The vertex of a parabola given by the equation $y = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$. In this case, $a = 1$. The vertex is given as $(2, -1)$, so the x-coordinate of the vertex is 2. Therefore, $2 = -\frac{b}{2(1)}$, which implies $2 = -\frac{b}{2}$. Multiplying both sides by 2 gives $4 = -b$, so $b = -4$. Alternatively, using the vertex form of the quadratic equation, $y = a(x - h)^2 + k$, where (h, k) is the vertex. Here, $a = 1$, $h = 2$, and $k = -1$. So, $y = 1(x - 2)^2 - 1$. Expanding this.

3 **[Foundational] • Target: Polynomial Remainder & Factor Theorems**

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(1, 4)$, and it passes through the point $(-1, 0)$. Find the value of b.

Final Answer: 2. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. Given the vertex is $(1, 4)$, we have $f(x) = a(x - 1)^2 + 4$. The parabola passes through the point $(-1, 0)$. Substituting these coordinates into the equation: $0 = a(-1 - 1)^2 + 4$ $0 = a(-2)^2 + 4$ $0 = 4a + 4$ $4a = -4$ $a = -1$. So, the function is $f(x) = -1(x - 1)^2 + 4$. Expanding this: $f(x) = -(x^2 - 2x + 1) + 4$ $f(x) = -x^2 + 2x - 1 + 4$ $f(x) = -x^2 + 2x + 3$. Comparing this to the general form.

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4 [Foundational] • Target: Transformations of Modulus Graphs

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx - 2$. The graph passes through the points $(1, -3)$ and $(2, 2)$. Determine the value of a.

Final Answer: 2. The function is given by $f(x) = ax^2 + bx - 2$. Since the graph passes through $(1, -3)$, we have $f(1) = -3$: $a(1)^2 + b(1) - 2 = -3$ $a + b - 2 = -3$ $a + b = -1$ (Equation 1) Since the graph passes through $(2, 2)$, we have $f(2) = 2$: $a(2)^2 + b(2) - 2 = 2$ $4a + 2b - 2 = 2$ $4a + 2b = 4$ $2a + b = 2$ (Equation 2) Now we solve the system of linear equations for a and b. Subtract Equation 1 from Equation 2: $(2a + b) - (a + b) = 2 - (-1)$ $a = 3$ Substitute $a = 3$ into Equation 1.

5 [Foundational] • Target: Solving Modulus Equations

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is a parabola. Find the x-coordinates of the points where the graph intersects the x-axis.

Final Answer: 1 and 2. To find the x-coordinates of the points where the graph intersects the x-axis, we need to find the values of x for which $f(x) = 0$. Set the function equal to zero: $x^2 - 3x + 2 = 0$ Factor the quadratic equation: $(x - 1)(x - 2) = 0$ This gives two possible solutions for x: $x - 1 = 0 \Rightarrow x = 1$ $x - 2 = 0 \Rightarrow x = 2$ Therefore, the graph intersects the x-axis at $x = 1$ and $x = 2$.

6 [Foundational] • Target: Modulus Inequalities

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. The vertex of the parabola is at point C. Determine the coordinates of the point where the graph intersects the y-axis.

Final Answer: (0, 3). To find the y-intercept of the graph of a function, we set $x = 0$ in the function's equation. For the function $f(x) = x^2 - 4x + 3$, substituting $x = 0$ gives: $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$ Therefore, the graph intersects the y-axis at the point (0, 3).

7 [Foundational] • Target: Polynomial Remainder & Factor Theorems

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is shown. Determine the y-coordinate of the y-intercept.

Final Answer: 2. The y-intercept occurs when $x = 0$. Substituting $x = 0$ into the function: $f(0) = (0)^2 - 3(0) + 2 = 0 - 0 + 2 = 2$. Therefore, the y-coordinate of the y-intercept is 2.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Transformations of Modulus Graphs**

Consider the function $f(x) = \frac{ax^2+bx+c}{dx+e}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = 3$. The graph also passes through the points $(0, 0)$ and $(1, -3)$. Find the value of $\frac{a}{d}$.

Final Answer: 3. The vertical asymptote at $x = 2$ implies that the denominator $dx + e$ is zero when $x = 2$, so $2d + e = 0$, which means $e = -2d$. Thus, the denominator can be written as $d(x - 2)$. The function becomes $f(x) = \frac{ax^2+bx+c}{d(x-2)}$. The horizontal asymptote at $y = 3$ implies that the limit of $f(x)$ as $x \to \pm\infty$ is 3. For a rational function, this limit is the ratio of the leading coefficients if the degree of the numerator is equal to the degree of the denominator.

9 **[Intermediate]** • **Target: Solving Modulus Equations**

Consider the function $f(x) = \frac{ax^2+bx+c}{x-d}$, where $a, b, c, d \in \mathbb{R}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 3$ and a horizontal asymptote at $y = 2$. The function passes through the point $(1, -8)$. If $f'(x)$ has a critical point at $x = 5$, find the value of d.

The vertical asymptote at $x = 3$ implies that the denominator is zero at $x = 3$, so $d = 3$. The horizontal asymptote $y = 2$ for a rational function where the degree of the numerator is equal to the degree of the denominator is the ratio of the leading coefficients. Therefore, $\frac{a}{1} = 2$, which means $a = 2$. Now we have $f(x) = \frac{2x^2+bx+c}{x-3}$. The function passes through $(1, -8)$: $-8 = \frac{2(1)^2+b(1)+c}{1-3} = \frac{2+b+c}{-2}$ $16 = 2 + b + c \Rightarrow b + c = 14$. Next, we find the.

10 **[Intermediate]** • **Target: Modulus Inequalities**

Determine the equation of the oblique asymptote to the graph of $f(x) = \frac{2x^2-3x+4}{x-1}$.

Final Answer: $y = 2x - 1$. To find the equation of the oblique asymptote, divide the numerator by the denominator using polynomial long division or synthetic division:

$2x^2 - 3x + 4, = 2x(x-1) - x + 4;, = 2x(x-1) - 1(x-1) + 3;, = (2x-1)(x-1) + 3$ Dividing through by $x - 1$ yields: $f(x) = 2x - 1 + \frac{3}{x-1}$ As $x \to \pm\infty$, the remainder term $\frac{3}{x-1} \to 0$. Therefore, the linear quotient gives the oblique asymptote: $y = 2x - 1$ - $y = 2x - 3$ arises from mista.

11 **[Intermediate]** • **Target: Polynomial Remainder & Factor Theorems**

Evaluate $\cos\left(\sin^{-1}\left(-\frac{1}{2}\right)\right)$.

Final Answer: $\sqrt{\frac{3}{4}}$. Working through this functions step: Let $\theta = \sin^{-1}(-\frac{1}{2}),$ so $\sin\theta = -\frac{1}{2}.$ Then $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{\frac{3}{4}}.$

12 **[Intermediate]** • **Target: Transformations of Modulus Graphs**

Evaluate $\cos\left(\sin^{-1}\left(\frac{1}{2}\right)\right)$.

Final Answer: $\sqrt{\frac{3}{4}}$. Working through this functions step: Let $\theta = \sin^{-1}(\frac{1}{2}),$ so $\sin\theta = \frac{1}{2}.$ Then $\cos\theta = \sqrt{1-\sin^2\theta} = \sqrt{\frac{3}{4}}.$

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13 **[Intermediate]** • **Target: Solving Modulus Equations**

Find the oblique asymptote of $f(x) = \frac{4x^2 + 3x - 14}{(x + 2)}$, in the form $y = mx + c$.

Final Answer: $y = 4x - 5$. Working through this functions step:

Dividing : $\frac{4x^2 + 3x - 14}{(x + 2)} = 4x - 5 + \frac{-4}{(x + 2)}$, so the asymptote is $y = 4x - 5$.

14 **[Intermediate]** • **Target: Modulus Inequalities**

Find the oblique asymptote of $f(x) = \frac{3x^2 - 10x + 11}{(x - 2)}$, in the form $y = mx + c$.

Final Answer: $y = 3x - 4$. Working through this functions step:

Dividing : $\frac{3x^2 - 10x + 11}{(x - 2)} = 3x - 4 + \frac{3}{(x - 2)}$, so the asymptote is $y = 3x - 4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Polynomial Remainder & Factor Theorems** • **Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity**

Find the oblique asymptote of $f(x) = \frac{2x^2 + 4x - 20}{(x + 4)}$, in the form $y = mx + c$.

Final Answer: $y = 2x - 4$. Working through this functions step:

Dividing : $\frac{2x^2 + 4x - 20}{(x + 4)} = 2x - 4 + \frac{-4}{(x + 4)}$, so the asymptote is $y = 2x - 4$.

16 **[Challenging]** • **Target: Transformations of Modulus Graphs** • **Cross-Lesson: 1.04 Partial Fractions**

Find the oblique asymptote of $f(x) = \frac{4x^2 - 11x + 2}{(x - 2)}$, in the form $y = mx + c$.

Final Answer: $y = 4x - 3$. Working through this functions step:

Dividing : $\frac{4x^2 - 11x + 2}{(x - 2)} = 4x - 3 + \frac{-4}{(x - 2)}$, so the asymptote is $y = 4x - 3$.

17 **[Challenging]** • **Target: Solving Modulus Equations** • **Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)**

Find the oblique asymptote of $f(x) = \frac{4x^2 - 14x + 7}{(x - 3)}$, in the form $y = mx + c$.

Final Answer: $y = 4x - 2$. Working through this functions step:

Dividing : $\frac{4x^2 - 14x + 7}{(x - 3)} = 4x - 2 + \frac{1}{(x - 3)}$, so the asymptote is $y = 4x - 2$.

18 **[Challenging]** • **Target: Modulus Inequalities** • **Cross-Lesson: 3.01 3D Vectors & the Cross Product**

Find the oblique asymptote of $f(x) = \frac{4x^2 + 19x - 2}{(x + 5)}$, in the form $y = mx + c$.

Final Answer: $y = 4x - 1$. Working through this functions step:

Dividing : $\frac{4x^2 + 19x - 2}{(x + 5)} = 4x - 1 + \frac{3}{(x + 5)}$, so the asymptote is $y = 4x - 1$.

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19 [Challenging] • Target: Polynomial Remainder & Factor Theorems • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

Find the oblique asymptote of $f(x) = \frac{2x^2 + 9x + 8}{(x + 4)}$, in the form $y = mx + c$.

Final Answer: $y = 2x + 1$. Working through this functions step:

Dividing : $\frac{2x^2 + 9x + 8}{(x + 4)} = 2x + 1 + \frac{4}{(x + 4)}$, so the asymptote is $y = 2x + 1$.

20 [Challenging] • Target: Transformations of Modulus Graphs • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Find the oblique asymptote of $f(x) = \frac{2x^2 - 4x + 3}{(x - 1)}$, in the form $y = mx + c$.

Final Answer: $y = 2x - 2$. Working through this functions step:

Dividing : $\frac{2x^2 - 4x + 3}{(x - 1)} = 2x - 2 + \frac{1}{(x - 1)}$, so the asymptote is $y = 2x - 2$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Solving Modulus Equations

Solve the equation $|2x - 5| = x + 1$.

a. $x = 6$ **and** $x = \frac{4}{3}$ ✓

b. $x = 6$ only

c. $x = \frac{4}{3}$ only

d. $x = -6$ and $x = -\frac{4}{3}$

Case 1: $2x - 5 = x + 1 \Rightarrow x = 6$. Check: $|2(6) - 5| = 7 = 6 + 1$. Valid. Case 2: $2x - 5 = -(x + 1) = -x - 1 \Rightarrow 3x = 4 \Rightarrow x = 4/3$. Check: $|8/3 - 5| = 7/3 = 4/3 + 1$. Both are valid.

22 [Foundational] • Target: Solving Modulus Equations

Solve $|3x + 2| = |x - 4|$.

a. $x = \frac{1}{2}$ **and** $x = -3$ ✓

b. $x = 2$ and $x = -3$

c. $x = -\frac{1}{2}$ and $x = 3$

d. $x = 1$ and $x = -4$

Square both sides or branch: $3x + 2 = \pm(x - 4)$. Branch 1: $3x + 2 = x - 4 \Rightarrow 2x = -6 \Rightarrow x = -3$. Branch 2: $3x + 2 = -x + 4 \Rightarrow 4x = 2 \Rightarrow x = 1/2$.

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23 [Foundational] • Target: Solving Modulus Equations

Why can $|x - 7| = -3$ have no real solutions?

a. The absolute value of any real expression is non-negative ($|u| \geq 0$), so it cannot equal a negative number ✓

b. Because x must be greater than 7

c. Because -3 is not a prime number

d. The equation has complex solutions only

By definition, $|u| \geq 0$ for all $u \in \mathbb{R}$. A non-negative quantity can never equal -3.

24 [Foundational] • Target: Modulus Inequalities

Solve the inequality $|2x - 3| < 7$.

a. $-2 < x < 5$ ✓

b. $x < 5$

c. $x < -2$ or $x > 5$

d. $-5 < x < 5$

$|2x - 3| < 7 \Leftrightarrow -7 < 2x - 3 < 7$. Add 3: $-4 < 2x < 10$. Divide by 2: $-2 < x < 5$.

25 [Foundational] • Target: Modulus Inequalities

Solve $|3x + 1| \geq 10$.

a. $x \leq -\frac{11}{3}$ **or** $x \geq 3$ ✓

b. $-\frac{11}{3} \leq x \leq 3$

c. $x \geq 3$

d. $x \leq -3$ or $x \geq \frac{11}{3}$

$|u| \geq a \Leftrightarrow u \leq -a$ or $u \geq a$. Branch 1: $3x + 1 \leq -10 \Rightarrow 3x \leq -11 \Rightarrow x \leq -11/3$. Branch 2: $3x + 1 \geq 10 \Rightarrow 3x \geq 9 \Rightarrow x \geq 3$.

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26 [Foundational] • Target: Modulus Inequalities

Which interval notation represents the solution to $|x - a| \leq \varepsilon$ where $\varepsilon > 0$?

a. $[a - \varepsilon, a + \varepsilon]$ ✓

b. $(a - \varepsilon, a + \varepsilon)$

c. $[-\varepsilon, \varepsilon]$

d. $[a, a + \varepsilon]$

$|x - a| \leq \varepsilon \Leftrightarrow -\varepsilon \leq x - a \leq \varepsilon \Leftrightarrow a - \varepsilon \leq x \leq a + \varepsilon$, giving $[a - \varepsilon, a + \varepsilon]$.

27 [Foundational] • Target: Polynomial Remainder & Factor Theorems

When $P(x) = 2x^3 - 3x^2 + ax - 5$ is divided by $(x - 2)$, the remainder is 11. Find a.

a. 6 ✓

b. 4

c. -6

d. 8

By the Remainder Theorem, $P(2) = 11$. $P(2) = 2(8) - 3(4) + 2a - 5 = 16 - 12 + 2a - 5 = 2a - 1$. Setting $2a - 1 = 11 \Rightarrow 2a = 12 \Rightarrow a = 6$.

28 [Foundational] • Target: Polynomial Remainder & Factor Theorems

If $(x + 3)$ is a factor of $Q(x) = x^3 + kx^2 - 7x + 6$, find k.

a. 2 ✓

b. -2

c. 3

d. -1

By the Factor Theorem, $Q(-3) = 0$. $Q(-3) = (-3)^3 + k(-3)^2 - 7(-3) + 6 = -27 + 9k + 21 + 6 = 9k$. Setting $9k = 0 \Rightarrow k = 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Polynomial Remainder & Factor Theorems

A polynomial $f(x)$ leaves remainder 3 when divided by $(x - 1)$ and remainder 7 when divided by $(x - 3)$. What is the linear remainder when $f(x)$ is divided by $(x - 1)(x - 3)$?

a. $2x + 1$ ✓

b. $2x - 1$

c. $x + 2$

d. $4x - 1$

Let $R(x) = ax + b$. $R(1) = a + b = 3$ and $R(3) = 3a + b = 7$. Subtracting: $2a = 4 \Rightarrow a = 2$. Then $b = 3 - 2 = 1$. The remainder is $2x + 1$.

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30 [Intermediate] • Target: Transformations of Modulus Graphs

How is the graph of $y = |f(x)|$ obtained from the graph of $y = f(x)$?

a. Any part of the graph lying below the x-axis is reflected vertically across the x-axis into the upper half-plane ✓

b. The left half of the graph is discarded and replaced by reflection of the right half

c. The graph is shifted 1 unit upward

d. The graph is reflected across the y-axis

Taking $|f(x)|$ preserves positive y-values and takes the opposite of negative y-values, reflecting sections below the x-axis upward across the x-axis.

31 [Intermediate] • Target: Transformations of Modulus Graphs

How is the graph of $y = f(|x|)$ obtained from $y = f(x)$?

a. Discard the portion for $x < 0$, and reflect the portion for $x \geq 0$ across the y-axis to form an even function ✓

b. Reflect the entire graph across the x-axis

c. Reflect the graph across $y = x$

d. Shift the graph to the right by $|x|$

Because $|-x| = |x|$, $f(|x|)$ is an even function: the right half ($x \geq 0$) is reflected across the y-axis, discarding the original $x < 0$ part.

32 [Intermediate] • Target: Transformations of Modulus Graphs

What is the vertex (sharp turn) of the function $g(x) = 3|x - 4| - 5$?

a. $(4, -5)$ ✓

b. $(-4, -5)$

c. $(4, 5)$

d. $(0, -5)$

The absolute value term $|x - 4|$ reaches its minimum of 0 at $x = 4$. At $x = 4$, $g(4) = 0 - 5 = -5$, giving vertex $(4, -5)$.

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33 [Intermediate] • Target: Solving Modulus Equations

Find the oblique asymptote of $f(x) = \frac{x^2 - 14}{(x + 4)}$, in the form $y = mx + c$.

a. $y = 2x - 5$

b. $y = 3x - 5$

c. $y = x - 4$ ✓

d. $y = 3x - 2$

Working through this functions step: Dividing : $\frac{x^2 - 14}{(x + 4)} = x - 4 + \frac{2}{(x + 4)}$, so the asymptote is $y = x - 4$.

34 [Intermediate] • Target: Modulus Inequalities

Find the oblique asymptote of $f(x) = \frac{4x^2 + 3x - 3}{(x)}$, in the form $y = mx + c$.

a. $y = 6x + 4$

b. $y = 4x + 3$ ✓

c. $y = 3x + 5$

d. $y = 6x + 1$

Working through this functions step: Dividing : $\frac{4x^2 + 3x - 3}{(x)} = 4x + 3 + \frac{-3}{(x)}$, so the asymptote is $y = 4x + 3$.

35 [Intermediate] • Target: Polynomial Remainder & Factor Theorems

Find the oblique asymptote of $f(x) = \frac{x^2 + 1}{(x + 1)}$, in the form $y = mx + c$.

a. $y = x - 1$ ✓

b. $y = 2x + 1$

c. $y = 3x$

d. $y = 2x - 2$

Working through this functions step: Dividing : $\frac{x^2 + 1}{(x + 1)} = x - 1 + \frac{2}{(x + 1)}$, so the asymptote is $y = x - 1$.

36 [Intermediate] • Target: Transformations of Modulus Graphs

Find the oblique asymptote of $f(x) = \frac{x^2 - 2}{(x - 2)}$, in the form $y = mx + c$.

a. $y = 3x + 3$

b. $y = 2x$

c. $y = x + 2$ ✓

d. $y = 2x + 4$

Working through this functions step: Dividing : $\frac{x^2 - 2}{(x - 2)} = x + 2 + \frac{2}{(x - 2)}$, so the asymptote is $y = x + 2$.

37 [Intermediate] • Target: Solving Modulus Equations

Evaluate $\cos\left(\sin^{-1}(-1)\right)$.

a. -1

b. 0 ✓

c. $0 + 3$

d. 2

Working through this functions step: Let $\theta = \sin^{-1}(-1)$, so $\sin\theta = -1$. Then $\cos\theta = \sqrt{1 - \sin^2\theta} = 0$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 **[Challenging]** • **Target: Modulus Inequalities** • **Cross-Lesson: 3.01 3D Vectors & the Cross Product**

Evaluate $\cos\left(\sin^{-1}(0)\right)$.

a. 1 ✓

b. 0

c. $1+3$

d. -1

Working through this functions step: Let $\theta = \sin^{-1}(0)$, so $\sin\theta = 0$. Then $\cos\theta = \sqrt{1-\sin^2\theta} = 1$.

39 **[Challenging]** • **Target: Polynomial Remainder & Factor Theorems** • **Cross-Lesson: 3.02 Lines and Planes in Three Dimensions**

Evaluate $\cos\left(\sin^{-1}(1)\right)$.

a. -1

b. 0 ✓

c. 1

d. $0+3$

Working through this functions step: Let $\theta = \sin^{-1}(1)$, so $\sin\theta = 1$. Then $\cos\theta = \sqrt{1-\sin^2\theta} = 0$.

40 **[Challenging]** • **Target: Transformations of Modulus Graphs** • **Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)**

Solve $|x+1| = |-3x-1|$. Give the largest solution.

a. -2

b. 4

c. 3

d. **0** ✓

Working through this functions step: Case 1: $x+1 = -3x-1$. Case 2: $x+1 = -(-3x-1)$. Largest solution: $x = 0$

41 **[Challenging]** • **Target: Solving Modulus Equations** • **Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals**

Solve $|2x+5| = |3x+3|$. Give the largest solution.

a. 6

b. -2

c. 1

d. **2** ✓

Working through this functions step: Case 1: $2x+5 = 3x+3$. Case 2: $2x+5 = -(3x+3)$. Largest solution: $x = 2$

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42 [Challenging] • Target: Modulus Inequalities • Cross-Lesson: 4.02 The Poisson Distribution

Solve $|x + 4| = |-2x - 2|$. Give the largest solution.

a. **2** ✓

b. 5

c. 1

d. 4

Working through this functions step: Case 1: $x + 4 = -2x - 2$. Case 2: $x + 4 = -(-2x - 2)$. Largest solution: $x = 2$

43 [Challenging] • Target: Polynomial Remainder & Factor Theorems • Cross-Lesson: 5.01 Further Integration Techniques

Solve $|x + 3| = |2x + 6|$. Give the largest solution.

a. -1

b. -7

c. **-3** ✓

d. 0

Working through this functions step: Case 1: $x + 3 = 2x + 6$. Case 2: $x + 3 = -(2x + 6)$. Largest solution: $x = -3$

44 [Challenging] • Target: Transformations of Modulus Graphs • Cross-Lesson: 5.02 Differential Equations

Solve $|x + 3| = |-x + 1|$. Give the largest solution.

a. **-1** ✓

b. 0

c. -2

d. -5

Working through this functions step: Case 1: $x + 3 = -x + 1$. Case 2: $x + 3 = -(-x + 1)$. Largest solution: $x = -1$

45 [Challenging] • Target: Solving Modulus Equations • Cross-Lesson: 5.03 Maclaurin Series & Convergence of Series

Solve $|x - 3| = |-x - 5|$. Give the largest solution.

a. -5

b. 0

c. -2

d. **-1** ✓

Working through this functions step: Case 1: $x - 3 = -x - 5$. Case 2: $x - 3 = -(-x - 5)$. Largest solution: $x = -1$

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